

An LLM Agent for the Selection, Validation and Interpretation of Statistical Methods

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1. Before you start

Sixteen slides, about 20 minutes when spoken at a normal pace. If time is short, move quickly through slides 10 and 11 and keep slides 2, 9 and 13 at full length. Those three carry the problem, the honest reading of the results, and the evidence that the measurement itself was checked.

What a professor listens for is not a large number. It is whether you know exactly what your evidence supports, and whether you say where it stops before being asked.

2. The problem statement, at three lengths

In one sentence

“Language models produce statistical analyses that are fluent and arithmetically correct but often invalid, because validity depends on how the data was collected — which the data itself does not contain.”

In thirty seconds

“Ask an AI to analyse a dataset and it will always give you an answer. It reads well and the arithmetic is right. But whether the method is valid depends on facts about the study design — are the observations independent, paired, ordered in time — and none of that is visible in the spreadsheet. So the model can apply a method the data does not support, invent a statistic, or answer when it should decline. This project tests whether giving the model an explicit decision procedure fixes that.”

In two minutes

- Open with the observation: an LLM always answers, and its answers are fluent and arithmetically correct.
- Name the gap: validity is decided by design facts — independence, pairing, temporal order, clustering — that live outside the data file.
- Give the three consequences: an invalid method with correct numbers; a fabricated statistic that reads as a result; no obligation to decline.
- Say why it matters: the people most likely to rely on an AI's analysis are the people least able to audit its method.

- State the question: does an explicit protocol make the model choose valid methods more reliably than tool access alone?

3. Slide by slide

Slide 1 · Title

Establish who you are and the single question. (about 0.5 min)

- Introduce yourself and the project title.
- Read the question on the slide: does an explicit decision protocol make an LLM choose valid statistical methods more reliably than tool access alone?

Land this: Keep this short. The next slide does the real work.

Transition: “Let me start with why this question matters.”

Slide 2 · Problem statement

The most important slide. Everything after it answers it. (about 2 min)

- State the problem in one breath: LLM analyses are fluent, arithmetically correct, and frequently invalid.
- Explain why: validity depends on how the data was collected — independence, pairing, time order, clustering — and those facts are not in the data file.
- Walk the three consequences: an invalid method with correct arithmetic; a fabricated number that reads like a result; no mechanism to decline.

Land this: “The data itself does not contain the facts that decide whether the method is valid.”

Transition: “Here is what that looks like on real data.”

Slide 3 · A motivating case from the benchmark

Make the abstract problem concrete with one dataset. (about 1.5 min)

- Seoul Bike Sharing: 8,760 rows, one per hour for a year.
- The table looks like a textbook correlation or Poisson regression; both assume independent observations.
- Adjacent hours share weather, commuting and daylight, so they are not independent — and nothing in the file says so. The design card supplies that fact.

Land this: “The table looked perfect, and the correct answer was to decline.”

Transition: “So the question I tested is this.”

Slide 4 · Research question and hypotheses

Show exactly what is being tested and against what. (about 1 min)

- Read the research question; then H1 selection, H2 abstention, H3 provenance.
- Stress that H1 compares C with B, not A. B has the same tools and report format, so it is the honest control.

Land this: “If C beats B, it is the procedure that won — not better tools.”

Transition: “First, the design decisions that make this testable.”

Slide 5 · Scope and design decisions

Justify the closed library and the absence of code execution. (about 1 min)

- Fourteen methods across five task families, plus abstention.
- Closed so it is scorable; no general code execution, which would let the model bypass the policy.
- Unsupported designs are tested as abstention cases, not ignored.

Land this: “Declining is a decision, not a failure.”

Transition: “Here is how the system is built.”

Slide 6 · How it works: one shared core, one ordered protocol

Pre-empt the fairness question and show the intervention. (about 2 min)

- Top row: the three systems. Middle: the shared core — tool registry, result store, report schema, trace log. Only control flow differs.
- Bottom row: System C's seven steps. Point at steps 2 and 5 — design validation gates everything downstream, and step 5 permits declining.
- B can run every check too; it lacks the ordering and the permission to stop.

Land this: “The intervention is the order, not the tools.”

Transition: “One more design decision, about numbers.”

Slide 7 · The provenance contract

Explain how fabrication is prevented rather than detected. (about 1.25 min)

- Registered, rejected, referenced, resolved: results are stored under a key, raw numbers in prose are refused, references are substituted, and an unknown reference fails the run.
- Say the caveat yourself: for C this is a guarantee by design, not a finding. The empirical measure is the rejection rate.

Land this: “The model cannot write a number.”

Transition: “Now, how it was tested.”

Slide 8 · Benchmark and evaluation

Show the answer key is not opinion, and the test was not tuned. (about 2 min)

- Benchmark: 64 cases, 52 supported, 12 abstention, 48 held out; labels derived from the realised sample; four UCI datasets; the audit fixed one mislabel.
- Evaluation: frozen split, frozen prompts, two repetitions, 288 runs on Claude Haiku 4.5; McNemar's exact test on per-case majorities.

Land this: “48 of 64 labels are derived, not judged — and the test set was locked away.”

Transition: “So, the results.”

Slide 9 • Result: accuracy, and which differences are real

Deliver the headline, then separate what is established from what is not. (about 1.75 min)

- Left: A 75%, B 67%, C 89%. C over B is +25 points against a +10 target. B scored below A — tools alone did not help.
- Right: solid and filled means significant. C over B, $p = 0.0018$, established. C over A, +12.5, $p = 0.0703$, unresolved. B over A, not significant.

Land this: “A result I only believe when it agrees with me is not a result.”

Transition: “The safety results are even sharper.”

Slide 10 • Result: declining the invalid, and grounding the report

The two safety results. (about 1.25 min)

- Left: on design-hazard runs, C declined correctly 78% of the time; the tool-only baseline 0%. H2 supported.
- Right: the direct baseline had 43 of 96 reports rejected. B is at 100% because it uses real tools — its failure is the wrong method, not invented numbers. None reached a reader. H3 supported.

Land this: “That's exactly the failure this project set out to prevent.”

Transition: “Here's the mechanism behind it.”

Slide 11 • Result: where each system fails, and what C gave up

Show the mechanism, and answer the objection that C just refuses more. (about 1.25 min)

- Left: design-validation failures, B 20 against C 4. Totals A 24, B 32, C 11.
- Right: C answered 85% and was right on 87%; A and B answered 95% and 98% and were right less often.

Land this: “It declines the right cases, not at random.”

Transition: “Pulling the findings together.”

Slide 12 • Hypotheses revisited

Summarise against the questions asked. (about 0.5 min)

- H1, H2 and H3 supported.
- C over A unresolved — keep that row on the slide and say it.

Land this: “Three supported, one open.”

Transition: “Two things went wrong along the way, and they matter.”

Slide 13 · Two defects found in the measurement itself

Demonstrate that you checked your own instrument. (about 1.5 min)

- Defect one: 51 errored runs had already chosen a method, 37 correctly. Discarding them would have made the direct baseline look 6 points better.
- Defect two: the tie-break varied between sessions, so identical data gave p-values from 0.016 to 0.125. Fixed and enforced by a test.

Land this: “The system worked. The way I was measuring it did not, and nothing complained.”

Transition: “So here is what the evidence does not support.”

Slide 14 · Limitations and threats to validity

State the limits calmly and without apology. (about 1 min)

- Scoped to Claude Haiku 4.5; 48 cases and two repetitions resolve the main comparison but not C over A.
- Interpretation score withheld: the blinded round failed at kappa -0.044 .
- No independent second reviewer for public labels; some designs out of scope.

Land this: “Each limit is known, bounded and disclosed.”

Transition: “What the project contributes, and where it goes next.”

Slide 15 · Contributions and next steps

Close the argument. (about 1 min)

- Contributions: controlled evidence that the protocol drives valid selection; a benchmark that scores declining; provenance by construction.
- Next: a larger held-out set to resolve C over A, frontier models, a second scoring round, a wider method library.

Land this: “Structure, not tooling, makes an LLM a reliable statistician.” — then pause.

Transition: “You can see all of it running.”

Slide 16 · Thank you

Invite scrutiny. (about 0.5 min)

- Point to the live explorer and the repository.
- If there is time, open the Seoul bike case and show C declining while A and B both fit a model.

Land this: “Thank you — I’m happy to take questions.”

4. Numbers to know without looking

Every figure below matches the recorded held-out run. Quote them exactly; rounding a p-value in speech is where a sharp question starts.

Measure	Value
Selection accuracy	A 75.0% · B 66.7% · C 88.5%
C over B	+25.0 points · $p = 0.0018$ · 95% CI [+12.5, +39.6]
C over A	+12.5 points · $p = 0.0703$ · not established
Declined correctly	A 28% · B 0% · C 78% (design-hazard runs)
Reports rejected	A 43 of 96 · B 0 · C 8
Design-validation failures	B 20 · C 4
Selective accuracy	C 87% at 85% coverage
Benchmark	64 cases · 52 supported · 12 abstain · 48 held out
Runs	288 graded, 2 repetitions, Claude Haiku 4.5
Reliability failure	inter-rater kappa -0.044

5. Questions to expect

Answer the question asked, then stop. Where a limit is involved, state it plainly — defensiveness reads worse than the limit itself.

Q. If fabrication is impossible in System C, isn't its perfect fidelity a tautology?

A. Yes, and that is how it is reported. For C, fidelity is a property of the design, not a measured finding. The empirical measure is how often the model reached for a reference that did not exist, and that rate is reported.

Q. Why is the main comparison C against B rather than C against A?

A. B holds exactly the same tools and the same report format as C; only the control flow differs. Comparing C with B isolates the protocol. Comparing C with A would mix the protocol with tool access.

Q. C over A isn't significant. Is the protocol actually better than no tools?

A. The direction favours C — the discordant cases split seven to one — but at 48 cases that is not enough to rule out chance, so I report it as unresolved. The primary comparison, against B, is significant at $p = 0.0018$.

Q. How do you know your gold labels are correct?

A. 48 of 64 labels are construction-derived: design facts come from the generator and distributional facts from the sample actually drawn, so they are not judgements. The 16 public-data labels were audited against their realised values, which caught and fixed one mislabel.

Q. Why only two repetitions?

A. It was a resource constraint. Two repetitions are enough to establish the primary comparison and to measure run-to-run stability — 25 of 144 system-case cells disagreed between runs — but not to characterise the full distribution. More repetitions are the first item of future work.

Q. Why Claude Haiku 4.5 and not a frontier model?

A. Every held-out claim is scoped to Haiku 4.5, and I do not generalise beyond it. The runner is model-agnostic and the evaluation is frozen, so re-running on a larger model is a direct next step.

Q. Why didn't you set temperature to zero for determinism?

A. The temperature parameter has been removed on current models and returns an error if sent. Determinism is controlled instead by a pinned model, a fixed effort level and a frozen prompt hash — and residual variance is measured across repetitions rather than assumed away.

Q. The expert policy scores 100%. Doesn't that prove the system works?

A. No — that score is circular, because the expert policy is what derived the gold labels. It validates the harness, not the method. The informative row is the naive policy inside the same state machine, which scores far lower and never abstains — showing the architecture alone does not produce the gain.

Q. Why did you keep errored runs in the accuracy figure?

A. Because none of the 51 errors were infrastructure. Each run had already chosen a method before its report failed the provenance check, and 37 had chosen correctly. Discarding them would raise the direct baseline from 75.0% to 81.1% by hiding its worst behaviour. Runs that never reached a decision are excluded.

Q. Couldn't System B simply do what System C does?

A. It has every tool C has, so nothing stops it. What it lacks is the requirement to validate the design before anything else, and the explicit permission to decline. The results show that without those, it does not discover them.

Q. What happened with the interpretation scoring?

A. The blinded round failed its reliability check — inter-rater kappa of -0.044 , worse than chance. The rubric admitted two defensible readings of its middle grade, so it was rebuilt. I withheld the metric rather than report a number from an instrument that demonstrably does not measure anything.

Q. Why McNemar's test?

A. Each case is answered by every system, so outcomes are paired and binary. McNemar's exact test uses only the discordant cases, which is right for paired binary data with small discordant counts. The bootstrap gives an interval on the difference itself.

Q. What is the practical application?

A. Anywhere an LLM analyses data for people who cannot audit the method — business analysts, clinicians, policy teams. The risk is not bad arithmetic; it is a confident answer from a method the data does not support.

Q. What would you do differently?

A. A larger held-out set and more repetitions to resolve C over A, and an independent second reviewer for the public-data labels from the start.

6. Phrases to avoid

Instead of	Say
“The system proves LLMs can be trusted.”	“On this model and benchmark, the protocol beat tool access alone by 25 points.”
“It is better than a plain LLM.”	“It leads the plain LLM by 12.5 points, but that difference is not significant at this sample size.”
“Numerical fidelity is 100%.”	“Fabrication is prevented by design; the rejection rate is the empirical measure.”
“It never makes mistakes.”	“It fails least at design validation, which is the failure it was built to prevent.”
“The results show it works on all models.”	“The held-out result is scoped to Claude Haiku 4.5.”